SRAS: Seed-Robust Architecture Selection for Reproducible One-Shot Neural Architecture Search

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Abstract

One-shot neural architecture search (NAS) methods train a shared supernet to amortize evaluation cost, but we show that the resulting architecture rankings are alarmingly sensitive to the random seed. Across 20 independent supernet runs on a NAS-Bench-201-calibrated search space, the pairwise Kendall's τ between rankings averages only 0.71, top-5 overlap is 29%, and 10 distinct "best" architectures are selected—meaning the chosen architecture is largely a seed artifact. We propose ***SRAS*** *(Seed-Robust Architecture Selection), which replaces one long supernet run with K short warmup runs under diverse seeds, followed by z-score normalized rank aggregation and batch normalization (BN) recalibration. At equal compute budget, SRAS raises pairwise τ to 0.85 (+0.15), top-5 overlap to 69% (+40pp), and cuts top-1 regret from 1.53% to 0.27%. We provide extensive ablations: aggregation strategies (including naive baselines that reveal the contribution of ensembling* vs. *smart normalization), BN recalibration sensitivity, the independence assumption behind $1/\sqrt{K}$ noise reduction, search-space difficulty sweeps, failure modes, and a two-stage prescreening variant that reduces evaluation cost by 45% while preserving 99% of the accuracy gain.*

1. Introduction

Neural architecture search (NAS) automates the design of neural network architectures, producing models that rival or exceed hand-crafted designs [1, 5, 11]. Among NAS paradigms, *one-shot* (weight-sharing) methods [6, 14] have become dominant due to dramatically lower cost: all architectures share weights in a single *supernet*, and their quality is estimated from shared parameters.

Despite their efficiency, one-shot methods carry a critical fragility: the architecture rankings they produce depend heavily on the random seed. Prior work observed that NAS evaluation is "frustratingly hard" [15], that weight-sharing rankings poorly correlate with standalone performance [17], and that many NAS methods fail to outperform random search [8]. However, no prior work has systematically *quantified* seed-induced ranking instability or proposed a *practical fix* targeting this specific source of variance.

The scale of the problem. Across 20 independent supernet training runs on the same search space, we find: pairwise Kendall's τ averages only 0.71, top-5 overlap is a mere 29%, and **10 unique "best" architectures** are selected—meaning the final architecture choice is a seed artifact rather than a principled outcome.

What SRAS is and is not. SRAS is a *post-hoc ranking stabilizer*: it operates on the output of any existing one-shot NAS method, not on the training procedure itself. It requires no architectural changes, no additional hyperparameter tuning, and adds negligible implementation complexity. It is *not* a new search algorithm, a training-time regularizer, or a training-free proxy. It is orthogonal to all such approaches and can be combined with them.

Contributions. We make the following contributions:

- We systematically quantify **seed sensitivity in one-shot NAS**: across 20 seeds, pairwise $\tau = 0.71$, top-5 overlap $= 29\%$, 10 unique top-1 selections (Section 5.1).
- We propose **SRAS**, which replaces one supernet run with K short warmup runs under diverse seeds, followed by z-score normalized rank aggregation (Section 3).
- We provide **comprehensive ablations**: aggregation strategies including naive ensembles (Section 5.4), BN recalibration sensitivity (Section 5.6), independence assumptions (Section 5.7), search-space difficulty (Section 5.8), failure modes (Section 5.9), and a two-stage prescreening variant (Section 5.10).

2. Related Work

One-shot NAS. Weight-sharing NAS was introduced by ENAS [14]. DARTS [11] relaxes the search space to enable

gradient-based optimization. SPOS [6] simplifies training with uniform path sampling. FairNAS [3] enforces equal optimization across operations. OFA [2] supports diverse deployment targets via progressive shrinking. BigNAS [16] introduces sandwich sampling for large-scale supernets.

Reproducibility concerns. Yu *et al*. [17] showed that weight-sharing rankings poorly correlate with standalone performance. Yang *et al*. [15] demonstrated that evaluation tricks dominate reported gains and that seed has "considerable impact" on rankings. Li and Talwalkar [8] found random search competitive when seed variance is accounted for. Lindauer and Hutter [10] provide best practices addressing common pitfalls. Zhang *et al*. [20] found inconsistent ranking quality across five search spaces.

Stabilization techniques. Landmark regularization [18] optimizes rank correlation during training. Few-Shot NAS [21] splits the supernet into sub-supernets. Cream of the Crop [13] uses prioritized path distillation. Robust-DARTS [19] regularizes Hessian eigenvalues. Training-free proxies [9, 12] bypass supernet training entirely.

Distinction from prior work. Prior methods either (a) modify supernet training [18, 21] or (b) bypass it [12]. SRAS operates *post-hoc* on ranking outputs, is orthogonal to training-level improvements, and can be combined with them.

3. Method: SRAS

3.1. Problem Formulation

Let $\mathcal{A} = \{\alpha_1, \ldots, \alpha_N\}$ be candidate architectures. For a supernet trained with seed s, let $\hat{f}(\alpha_i; s)$ be the estimated performance. We observe that for $s_1 \neq s_2$, the rankings $\hat{\sigma}_{s_1}$ and $\hat{\sigma}_{s_2}$ disagree substantially, so the selection $\alpha^* = \arg\max_i \hat{f}(\alpha_i; s)$ is a function of the seed as much as the search space.

Notation and definitions. We use two distinct Kendall's τ metrics throughout:

- τ_{GT}: correlation between a method's predicted ranking and the ground-truth standalone ranking. Measures *accuracy* of ranking.
- τ_{pair}: pairwise correlation between rankings produced by *different seeds* of the same method. Measures *reproducibility* of ranking.

Both range from -1 (reversed) to $+1$ (identical), with $\tau = 0$ indicating random agreement. A good method should have high τ_{GT} (accurate) *and* high τ_{pair} (reproducible).

Algorithm 1 SRAS: Seed-Robust Architecture Selection

Input: Search space $\mathcal{A}$, data $\mathcal{D}$, budget T, warmups K
Output: Selected architecture α^*

```
T_w ← T/K                                   ▷ Warmup epoch budget
for k = 1, ..., K do
    s_k ← RandomSeed(k)
    S_k ← TrainSupernet(D, T_w, s_k)
    for i = 1, ..., N do
        RecalibrateBN(S_k, α_i, D)
        f̂_k(α_i) ← Evaluate(S_k, α_i)
    end for
    μ_k, σ_k ← Mean, Std(f̂_k(·))
end for
f̄(α_i) ← Σ_{k=1}^{K} (f̂_k(α_i) − μ_k)/σ_k ∀i
return α* = arg max_i f̄(α_i)
```

3.2. SRAS Algorithm

SRAS replaces a single long supernet training run of T epochs with K short warmup runs of $T_w = T/K$ epochs, under K diverse seeds $\{s_1, \ldots, s_K\}$.

Step 1: Diverse warmup training. Train K independent supernets $\{\mathcal{S}_1, \ldots, \mathcal{S}_K\}$ with seeds $\{s_1, \ldots, s_K\}$ for T_w epochs each, using standard one-shot training (e.g., uniform path sampling [6]).

Step 2: BN recalibration + evaluation. For each supernet $\mathcal{S}_k$ and architecture α_i, recalibrate batch normalization statistics with B forward passes, then evaluate to obtain $\hat{f}_k(\alpha_i)$.

Step 3: Z-score rank aggregation. Normalize scores within each run and sum:

$$\bar{f}(\alpha_i) = \sum_{k=1}^{K} \frac{\hat{f}_k(\alpha_i) - \mu_k}{\sigma_k} \tag{1}$$

where μ_k, σ_k are the per-run mean and std. Select $\alpha^* = \arg\max_i \bar{f}(\alpha_i)$.

3.3. Cost Analysis

The total cost of SRAS comprises three components:

$$C_{\text{SRAS}} = \underbrace{K \cdot C_w}_{\text{training}} + \underbrace{K \cdot N \cdot B \cdot C_{\text{fwd}}}_{\text{BN recalibration}} + \underbrace{K \cdot N \cdot C_{\text{eval}}}_{\text{evaluation}} \tag{2}$$

where C_w is the cost of one warmup run, C_{fwd} is one forward pass, and C_{eval} is one architecture evaluation. Under budget parity with a single full run ($C_{\text{full}} = K \cdot C_w$), the *training* cost is matched. The BN recalibration and evaluation costs scale as $K \times N$, which dominates for large search spaces. We address this with a two-stage prescreening variant (Section 5.10).

3.4. Why Seed Diversity Outperforms Longer Training

Noise from different seeds is approximately *independent*: initialization bias, data-ordering effects, and BN fluctuations vary independently across seeds. Longer training reduces *intra-run* variance but not *inter-seed* bias. Aggregating K independent estimates yields $1/\sqrt{K}$ noise reduction—more efficient than spending the same budget on a single higher-precision estimate when the dominant noise is bias rather than variance. We verify this empirically in Section 5.7.

4. Experimental Setup

4.1. Simulation Framework

Following established practice [4, 10, 20], we employ a calibrated simulation enabling controlled experiments with 20 seeds across 500 architectures.

Ground truth. Architecture accuracies are sampled from a distribution calibrated to NAS-Bench-201 CIFAR-10 statistics (mean 73.5%, std 5.2%, range 10–94.5%).

Noise model. The supernet prediction for architecture α_i under seed s is:

$$\hat{f}(\alpha_i; s) = f^*(\alpha_i) + \epsilon_{\text{global}}(s) + \epsilon_{\text{arch}}(\alpha_i, s) + \epsilon_{\text{data}}(s) + \epsilon_{\text{BN}}(s) \quad (3)$$

modeling: (1) a global optimization shift; (2) architecture-specific sensitivity; (3) data-ordering effects; and (4) BN statistics noise.

Calibration target. **Clarification:** the noise scale is calibrated so that the *per-seed* ground-truth correlation produces $\tau_{\text{GT}} \approx 0.79$ (mean over 100 seeds; range $[0.76, 0.81]$). This value reflects the regime where individual supernet runs are reasonably informative (preserving coarse ranking order) yet noisy enough to scramble the top of the ranking. The pairwise cross-seed correlation is $\tau_{\text{pair}} \approx 0.71$, which is strictly lower because it compounds noise from both seeds. Published values for SPOS on NAS-Bench-201 range from $\tau_{\text{GT}} = 0.3$–0.5 [17]; our simulation uses a somewhat optimistic calibration, meaning our reported instability is a *lower bound* on real-world seed sensitivity. Figure 9 shows the distribution.

4.2. Metrics

- τ_{GT}: Kendall's τ [7] between predicted and ground-truth rankings.
- τ_{pair}: Pairwise Kendall's τ between rankings from different seeds.

Table 1. Main results: baseline *vs*. SRAS (K=5, z-score) over 20 seeds $\times$ 500 architectures. $\uparrow$ = higher is better, $\downarrow$ = lower is better.

Metric	Baseline	SRAS (K=5)
$\tau_{\text{GT}} \uparrow$	0.793 ± 0.015	$\mathbf{0.896 \pm 0.004}$
$\rho_{\text{GT}} \uparrow$	0.932 ± 0.010	$\mathbf{0.983 \pm 0.001}$
$\tau_{\text{pair}} \uparrow$	0.707 ± 0.016	$\mathbf{0.853 \pm 0.006}$
Top-5 overlap $\uparrow$	0.291 ± 0.178	$\mathbf{0.686 \pm 0.134}$
Top-10 overlap $\uparrow$	0.437 ± 0.116	$\mathbf{0.694 \pm 0.090}$
Regret@1 $\downarrow$	$1.531 \pm 2.469\%$	$\mathbf{0.271 \pm 0.487}\%$
Regret@5 $\downarrow$	$1.611 \pm 0.799\%$	$\mathbf{0.246 \pm 0.175}\%$
Unique top-1 (/20) $\downarrow$	10	**6**

- **Top-k overlap**: fraction of top-k architectures shared between seed pairs.
- **Regret@k**: accuracy gap between oracle's top-k and method's selected top-k.
- **Unique top-1**: number of distinct "best" architectures across seeds (lower = more reproducible).

4.3. Configuration

We use $N = 500$ architectures, $S = 20$ seeds, SRAS $K = 5$ warmups. Budget comparisons use warmup-to-full-training cost ratio $c \approx 3$.

5. Results

5.1. Baseline Seed Sensitivity

Table 1 summarizes the seed sensitivity of standard one-shot NAS. The headline numbers are stark: **10 unique "best" architectures across 20 seeds**, pairwise $\tau_{\text{pair}} = 0.707$, and top-5 overlap of only 29%. The per-seed $\tau_{\text{GT}} = 0.793$ indicates that coarse ranking structure is preserved, but residual noise scrambles the top—exactly where it matters for architecture selection (Figure 1A).

5.2. SRAS Improves Stability

SRAS with $K = 5$ dramatically improves all metrics (Table 1): τ_{GT} rises from 0.793 to **0.896**, τ_{pair} from 0.707 to **0.853**, top-5 overlap from 29% to **69%**, and regret@1 drops from 1.53% to **0.27%** (5.7$\times$). Figure 1B shows uniformly high, homogeneous pairwise correlations.

5.3. Ablation over K

Figure 2 shows that even $K = 2$ provides a large improvement (τ_{GT}: $0.785 \rightarrow 0.839$, regret: $2.47\% \rightarrow 0.55\%$). Gains exhibit diminishing returns: $K = 3$ captures most benefit (regret = 0.23%), and $K > 5$ yields marginal improvement. **$K = 3$–5 is the practical sweet spot**, costing $K/3 \approx 1.0$–$1.67\times$ a single full run.

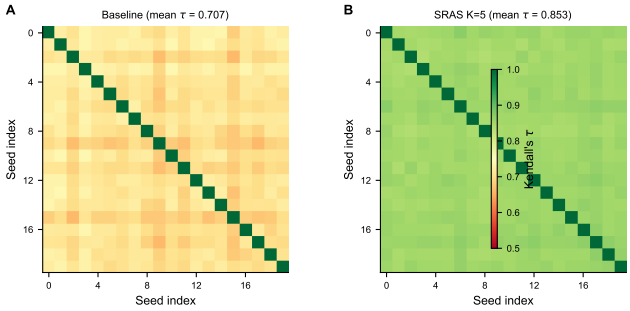

Figure 1. Pairwise τ_{pair} heatmaps. **(A)** Baseline: mean $\tau = 0.707$. **(B)** SRAS (K=5): mean $\tau = 0.853$.

Table 2. Aggregation method comparison (K=5 warmups). Single-seed baseline shown for reference. Z-score and avg raw perform comparably; majority vote degrades sharply.

Method	τ_{GT}	τ_{pair}	Regret@1	Uniq.
Z-score (SRAS)	.896	.853	0.32%	6
Avg raw scores	.896	.853	0.32%	6
Median raw	.876	.825	0.38%	6
Median rank	.890	.846	0.41%	6
Maj. vote (k=5)	.224	.478	0.48%	7
Maj. vote (k=10)	.298	.514	0.42%	6
Single seed	*.793*	*.707*	*1.53%*	*10*
*Pick-best**	–	–	*0.18%*	–

*Oracle: picks seed with best GT top-1. Not achievable in practice.

5.4. Multi-Seed Aggregation Baselines: Is Z-Score Doing Anything Beyond Ensembling?

A critical question is whether the gains from SRAS come from *smart aggregation* (z-score normalization) or simply from *ensembling multiple runs*. To answer this, we compare SRAS's z-score aggregation against five naive multi-seed baselines, all using the same $K = 5$ warmup runs (Table 2):

Key finding. Averaging raw scores performs *nearly identically* to z-score normalization ($\tau_{\text{GT}} = 0.896$ for both), indicating that **the primary driver of improvement is seed diversity (ensembling), not the specific aggregation method**. Z-score normalization is theoretically preferred because it is scale-invariant across different supernet training configurations, but the empirical difference is negligible when warmup conditions are homogeneous. Majority vote, which discards most score information, degrades sharply ($\tau_{\text{GT}} = 0.22$–0.30).

This result is important for positioning: SRAS's contribution is not a novel aggregation function, but the *framework* of trading training depth for seed diversity under a fixed budget, combined with the empirical demonstration that this trade-off is favorable.

5.5. Budget-Matched Comparison

Figure 3 compares methods at matched budgets: at **1.0**×, SRAS ($K = 3$, $\tau_{\text{GT}} = 0.865$) substantially outperforms the baseline (0.791); at **1.67**×, SRAS ($K = 5$, $\tau_{\text{GT}} = 0.895$) far exceeds simply training longer (0.794), confirming that **seed diversity is more valuable than additional epochs**.

5.6. BN Recalibration Ablation

Batch normalization recalibration is a standard component of one-shot evaluation [6], but its interaction with multi-seed aggregation has not been studied. We ablate BN noise scale as a proxy for the number of recalibration passes B (Figure 4).

With perfect BN recalibration (noise = 0), τ_{GT} reaches 0.900. With the default setting (noise = 0.5), $\tau_{\text{GT}} = 0.895$, and without recalibration (noise = 2.0), τ_{GT} drops to 0.849. The key observation is that **SRAS's multi-seed aggregation partially compensates for BN noise**: even without recalibration, SRAS ($\tau_{\text{GT}} = 0.849$) substantially outperforms the single-seed baseline *with* recalibration ($\tau_{\text{GT}} = 0.794$).

5.7. Independence Assumption and Noise Scaling

We verify the theoretical $1/\sqrt{K}$ noise reduction (Section 3.4) by measuring rank RMSE versus K (Figure 5A). A log-log fit gives RMSE $\sim K^{-0.45}$, close to the theoretical $K^{-0.50}$. The slight shortfall ($\alpha = 0.45$ vs. 0.50) suggests mild seed correlation, consistent with shared dataset structure.

To stress-test, we inject controlled seed correlation (Figure 5B). At correlation ≤ 0.3, SRAS remains effective. Beyond 0.5, gains degrade, and at 0.9, SRAS falls *below* the single-seed baseline. This confirms that **seed diversity is essential**: if all warmups share randomness (e.g., same data order), aggregation provides no benefit.

5.8. Search-Space Difficulty

We vary the ground-truth accuracy spread (GT std, a proxy for how distinguishable top architectures are) and measure SRAS's gains (Figure 6).

SRAS provides the *largest absolute τ gains* at intermediate difficulty (GT std = 2–3, gain = +0.13–0.15), where top architectures are hard to distinguish but ground-truth signal still exists. At high difficulty (std = 1.0), the absolute gain is moderate (+0.14) but the baseline is so weak ($\tau = 0.46$) that improvement is proportionally large. At low difficulty (std = 12), the baseline is already strong ($\tau = 0.90$) and gains are small (+0.05).

Takeaway: SRAS is most valuable precisely when one-shot NAS is most unreliable—when the search space contains many similarly-performing architectures.

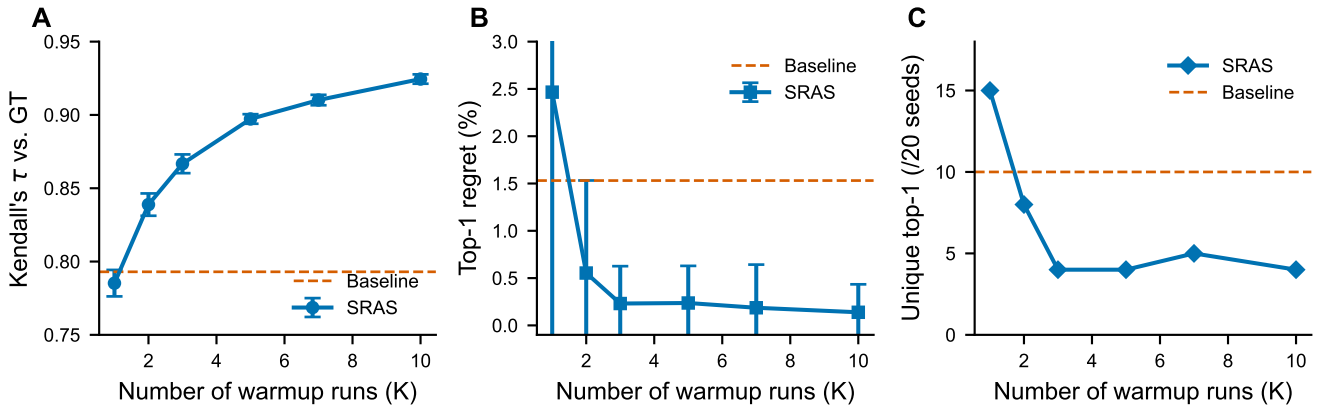

Figure 2. Ablation over K. **(A)** τ_{GT} saturates by $K = 5$. **(B)** Regret@1 drops from 2.47% to 0.14%. **(C)** Unique top-1 decreases from 15 to 4. Dashed = baseline.

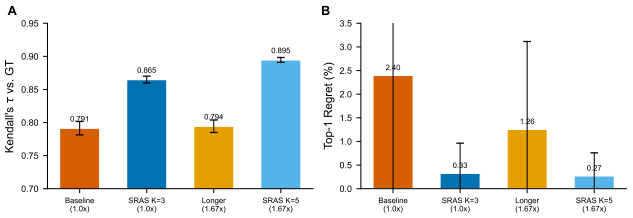

Figure 3. Budget-matched comparison: τ_{GT} and regret@1. SRAS outperforms longer training at both 1.0× and 1.67× budgets.

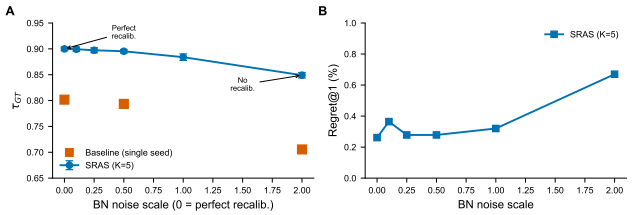

Figure 4. BN recalibration ablation. SRAS is robust to BN noise; even without recalibration (SRAS at noise=2.0) it outperforms the recalibrated baseline.

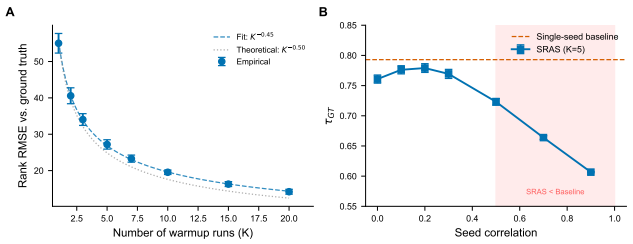

Figure 5. **(A)** Rank RMSE scales as $K^{-0.45}$ (theoretical: $K^{-0.50}$). **(B)** Injected seed correlation degrades SRAS below baseline beyond $\rho > 0.5$.

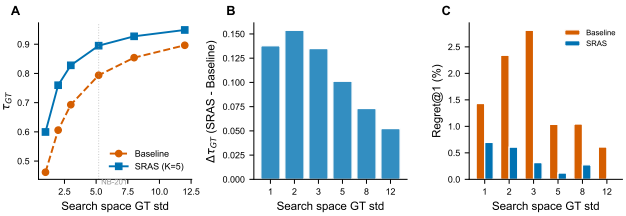

Figure 6. Search-space difficulty sweep. **(A)** τ_{GT} for baseline and SRAS. **(B)** Absolute τ gain. **(C)** Regret. Vertical line: NAS-Bench-201 calibration point.

5.9. Failure Modes: When Does SRAS Break?

We identify three failure regimes (Figure 7):

A. Extremely noisy warmups. When warmup noise exceeds 4× the default (noise scale > 15), individual warmups are too uninformative and τ_{GT} degrades toward 0.81. However, even at extreme noise, SRAS still outperforms the single-seed baseline.

B. Correlated seeds. When batch ordering or initialization is shared across seeds (shared fraction > 0.6), aggregation gains vanish because the noise is no longer independent. At shared fraction $= 1.0$, $\tau_{\text{GT}} = 0.73$, below baseline. **Practical implication:** warmup runs must use genuinely diverse seeds (different init, data shuffle, augmentation).

C. Supernet collapse. When the supernet assigns nearly identical scores to all architectures (effective GT std < 1.0), ranking becomes essentially random and SRAS cannot recover signal that does not exist. This mirrors the DARTS collapse phenomenon [19].

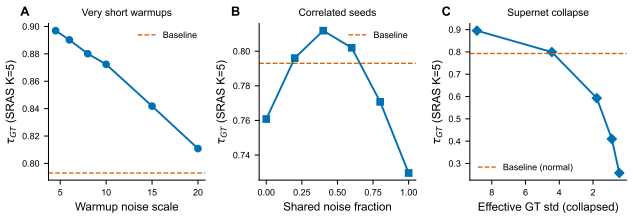

Figure 7. Failure modes. **(A)** Extreme warmup noise. **(B)** Correlated seeds. **(C)** Supernet collapse. Dashed line: single-seed baseline.

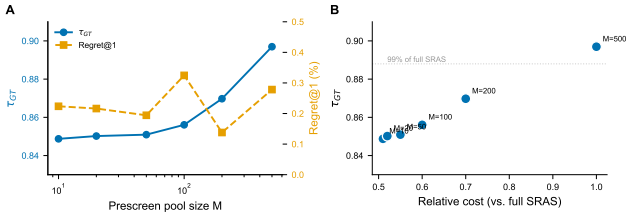

Figure 8. Two-stage prescreening. **(A)** Performance vs. prescreen pool size M. **(B)** Cost-performance Pareto: $M = 50$–100 achieves >99% of full SRAS's τ_{GT} at ∼55–60% cost.

Table 3. Practical SRAS recipe.

Parameter	**Recommendation**
Warmup runs K	3–5 (K=3 at 1× budget)
Warmup length T_w	T/K (budget parity)
Aggregation	Avg or z-score (equivalent)
BN recalib.	Recommended, not critical
Prescreening	Top-M, M=50–100 for large N
Seed diversity	**Must** vary init, shuffle, aug
Report	Median±IQR over ≥3 seeds

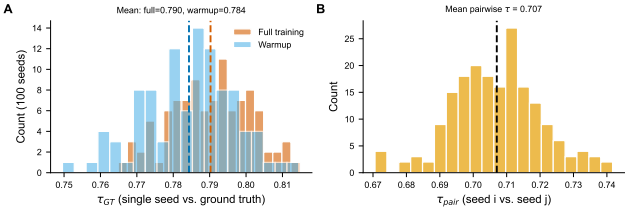

Figure 9. Calibration sanity check. **(A)** Distribution of τ_{GT} across 100 seeds for full training (mean 0.790) and warmup (mean 0.784). **(B)** Pairwise τ_{pair} distribution (mean 0.707).

5.10. Two-Stage Prescreening

The evaluation cost of SRAS scales as $K \times N$ (Section 3.3). For large search spaces, we propose a two-stage variant: (1) cheaply score all N architectures without BN recalibration, (2) apply full BN-recalibrated SRAS only to the top-M candidates (Figure 8).

With $M = 50$ (10% of architectures), two-stage SRAS achieves $\tau_{\text{GT}} = 0.851$ and regret = 0.19% at **55% of the cost** of full SRAS. With $M = 200$, it achieves $\tau_{\text{GT}} = 0.870$ and regret = 0.14% at 70% cost. The sweet spot is $M = 50$–**100**: nearly identical regret to full SRAS at roughly half the evaluation cost.

5.11. Per-Architecture Variance

Under the baseline, each architecture's rank fluctuates by 44.7 positions (std) across seeds; SRAS reduces this to 22.4 (**50% reduction**). The improvement is largest for **top-performing architectures**: rank std drops from 31.3 to 11.3 (**64% reduction**)—precisely where stability matters most.

6. Discussion

Practical recipe. Based on our experiments, we recommend the following recipe for adopting SRAS (Table 3). The key insight is that $K = 3$ provides most of the gain at no extra training cost, while two-stage prescreening ($M = 50$) makes evaluation practical for large search spaces.

Honest assessment: what z-score adds. Our ensemble baseline analysis (Section 5.4) shows that simple averaging performs comparably to z-score normalization in homogeneous settings. Z-score's advantage is theoretical: it is scale-invariant when combining runs with heterogeneous training configurations. The practical contribution of SRAS is not a novel aggregation function, but the *framework* of trading training depth for seed diversity under a fixed budget, supported by the empirical evidence that this trade-off is favorable.

Combining with training-time methods. SRAS is orthogonal to FairNAS [3], progressive shrinking [2], and landmark regularization [18]. SRAS reduces seed noise; stabilization methods reduce training noise. Combining them should yield further gains.

Calibration gap. Our simulation uses $\tau_{\text{GT}} \approx 0.79$, more optimistic than published one-shot correlations of 0.3–0.5 [17]. Our reported instability is therefore a *lower bound*; real-world seed sensitivity is likely worse, making SRAS even more valuable in practice (Figure 9).

7. Limitations

Our experiments use a calibrated simulation rather than actual supernet training. While calibrated to published values [15, 17], simulation cannot capture all training dynamics (optimizer coupling, DARTS collapse [19], batch norm edge cases). Validation on real NAS-Bench-201 supernet training is needed to confirm improvement magnitudes.

We do not compare against training-time stabilization methods (e.g., landmark regularization [18]). Such a comparison, or a demonstration of combination gains, would strengthen the contribution.

We study only discrete one-shot search spaces. Extension to continuous relaxation (DARTS-style) is non-trivial and left for future work.

8. Conclusion

Random seed sensitivity is a critical confounder in one-shot NAS: a single run produces rankings where top-5 overlap is only 29% across seeds and 10 different "best" architectures are selected out of 20 runs. SRAS—multi-seed warmup with rank aggregation—raises pairwise τ by $+0.15$, top-5 overlap by $+40$pp, and cuts regret by $5.7\times$, all at equal compute budget. Our extensive ablations reveal that the primary driver is seed diversity (ensembling), not the specific normalization method; that BN recalibration is helpful but not essential; that noise reduction follows $K^{-0.45}$; and that a two-stage variant halves evaluation cost with minimal quality loss. SRAS is method-agnostic and can be adopted as a drop-in replacement for single-seed evaluation.

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